

3. Real-Time Soil Moisture Mapping Using Scalable RF Sensor Networks (2025)

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Research Area: Intelligent Agricultural Monitoring and Automation System

Why it is Relevant

Supports the monitoring side of the rover by showing how intelligent systems collect field data. The paper demonstrates how scalable RF sensor networks can continuously monitor soil conditions and provide real-time information for agricultural decision-making.

Summary

This paper presents a real-time soil moisture mapping system based on scalable Radio Frequency (RF) sensor networks. Multiple RF sensing nodes are deployed across agricultural fields to continuously collect soil moisture data and transmit it wirelessly for processing. The system generates real-time moisture maps that help farmers understand field conditions, optimize irrigation schedules, and improve water-use efficiency. The scalable architecture enables large-area coverage while maintaining reliable communication and data collection. The results demonstrate the effectiveness of RF sensor networks for precision agriculture and intelligent field monitoring.

Relation to Your Project

- Real-time soil moisture monitoring
- RF-based sensing technology
- Wireless communication network
- Intelligent field data collection
- Precision irrigation management
- Agricultural automation
- Smart farming infrastructure
- Autonomous farming support